



Summary of this week

- This week, we studied the preliminary topics of particle filters.
- We reviewed key concepts and important definitions from the three prerequisite courses in this specialization.
- You learned how to express $f(x_k | \mathbb{Z}_k)$ as a two-step predict/update recursion.
- You were introduced to two ways to approximate an integral numerically: the rectangular and trapezoidal rules.
- You learned how to apply these rules to develop a brute-force approach to estimating the integrals involved in computing $f(x_k | \mathbb{Z}_k)$.
 - You were introduced to the “curse of dimensionality.”
- You learned how to implement this approach in Octave, and saw an example of doing so.



Where to from here?

- Next week, our focus shifts to finding a more efficient way to approximate an integral.
 - We are seeking to overcome the “curse of dimensionality” to find a more computationally efficient method.
- You will learn about the Monte–Carlo method for approximating an integral.
- You will also learn how to implement this Monte–Carlo method in Octave code.
- You will learn that a direct implementation of the Monte–Carlo method is not always possible, but that we can develop an indirect implementation using a surrogate “importance density.”
- You will also learn how to implement this modified Monte–Carlo method, using an importance density, in Octave code.



Credits

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